

Sensation

CT

Maintenance Instructions

System

Maintenance System

This document is valid for:

SOMATOM Sensation 10/16/Cardiac

SOMATOM Sensation 40/64/Cardiac 64

SOMATOM Sensation Open

The protocol CT02-023.832.01.08.02 is required for these instructions

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1 General

1.1 System maintenance intervals

Partial maintenance	every 6 months
Full maintenance	every 12 months
Preventive parts replacement	every 3 years

1.2 System maintenance times

1.2.1 Partial maintenance

Partial maintenance, performed every **6 months**, consists of the following parts:

Preparations	20 min
Preventive maintenance - part 2	90 min
Image quality check - constancy	30 min
Final steps	20 min

1.2.2 Full maintenance

Full maintenance, performed **every 12 months**, consists of the following parts:

Preparations	30 min
Safety inspections	60 min
Preventive maintenance - part 1	60 min
Preventive maintenance - part 2	90 min
Image quality check - constancy	30 min
Final steps	30 min

1.2.3 Preventive parts replacement

Preventive parts replacement, performed **every 3 years**, consists of the following parts:

Preventive parts replacement	30 min
------------------------------	--------

1.3 System maintenance planning

1.3.1 Partial maintenance

The partial maintenance is performed every 6 months and grouped into 3 packages as follows:

- Safety inspections
- Preventive maintenance - part 2
- Image quality check - constancy

1.3.2 Full maintenance

The full maintenance is performed every 12 months and includes the partial maintenance work as well. To improve planning and maintenance, the individual tasks have been grouped into 4 packages as follows :

- Safety inspections
- Preventive maintenance - part 1
- Preventive maintenance - part 2
- Image quality check - constancy

Each of these 4 packages may be performed independently of the others. Complete instructions are provided in the respective chapters. This makes it possible to schedule the full maintenance over several days. The work time required will be extended according to the respective 'Next steps' sections.

1.3.3 Preventive parts replacement

The UPS has to be replaced every 3 years beginning with the date of installation. The date for the next replacement has to be documented in the Maintenance Protocol at the time of the first maintenance and then at every subsequent maintenance interval.

1.4 Safety information

The installation, service and maintenance of equipment described herein is to be performed by qualified personnel who are employed by Siemens or one of its affiliates, or who are otherwise authorized by Siemens or one of its affiliates to provide such services. Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

In this context, "qualified" signifies that the service engineer has been trained and has had practical experience in the required routines so that he/she is capable of performing service work on the system. "Authorized" indicates that the service engineer is recognized by the system owner as a qualified service engineer and is therefore authorized to perform installation, service and, maintenance work on the system.

⚠ WARNING

[hz_serdoc_F13G01U12M02]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the "General safety notes" and the "Product-specific safety notes".
-

⚠ WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

- ⇒ The guidelines contained in this document have to be strictly followed.
-

1.4.1 Bloodborne pathogens

⚠ CAUTION

[hz_serdoc_F13G07U01M01]

Handling parts of the system that may have come into contact with patients may lead to infection through blood-borne pathogens.

Infection caused by blood-borne pathogens!

- ⇒ Take appropriate precautions against exposure to blood-borne pathogens (e.g. wear gloves).
-

1.4.2 High voltage

⚠ WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- ⇒ **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- ⇒ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- ⇒ **Ensure via measurements that all voltages are switched off.**
- ⇒ **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

1.4.3 Radiation protection



Observe the following protective measures when working on the CT unit with radiation "ON"

- use a radiation shield
- maintain a safe distance from the source
- use the minimum dose, smallest slice-thickness
- remove phantom (if possible)

⚠ CAUTION

[hz_serdoc_F13G01U03M01]

Not following X-rays protective regulations may lead to radiation exposure for yourself and/or other persons.

Risk of radiation exposure!

Observe radiation protection regulations when X-rays are switched on, e.g. :

- ⇒ **Never work inside the gantry room.**
- ⇒ **Do not leave the system unattended.**
- ⇒ **Make sure there is no one inside the gantry room.**
- ⇒ **Make sure that no person can enter the gantry room unnoticed (e.g. lock doors if necessary) .**

CAUTION

[hz_serdoc_F13G01U03M02]

Certain tests and procedures use X-rays. Observe radiation protection regulations (CT basic course).

Risk of exposure to X-rays.

- ⇒ **It is prohibited to perform service on the system, if for any reason the appropriate training or safety instructions regarding X-ray protection were not received during the CT training or through other qualified sources.**

1.4.4 Working in the gantry

Prior to starting work in the gantry:



1. Switch the system off; observe the power-off instructions and wait times as described above.
2. Wait until rotation has stopped.
3. Disable gantry rotation via service switch S301
4. Always use the rotation safety bolt to lock the rotation part of the gantry.
5. Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry. Use protective conductor cable 4665329 B1945

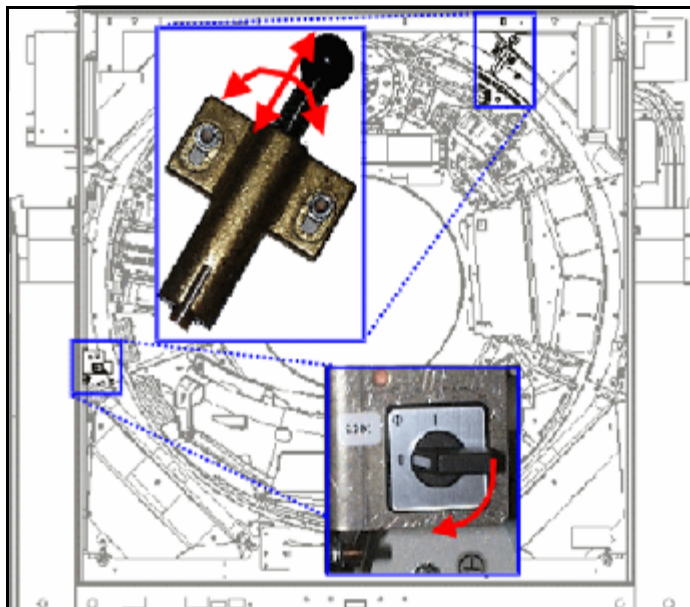


Fig. 1: Location of rotation safety devices

NOTE

Remove the cable and rotation safety bolt prior to rotating the gantry and enable gantry rotation (S301) after finishing work.

2 Prerequisites

2.1 Required tools and auxiliary materials

2.1.1 Documents

- Latest version of system **service documentation CT02-023.801.01** xx.xx
- **Maintenance Protocol** CT02-023.832.01. xx.xx **in printed form** .

2.1.2 Tools

- Standard tool kit
- Sliding calipers
- Torque wrench (3 Nm, 6 Nm, 28 Nm)
- Vacuum cleaner

2.1.3 Measurement devices and auxiliary materials

- Lubricant gun
- Tape measure
- Protective conductor meter (e.g. item number 44 158 99 RV090 or SECUTEST SIII))
- Service (termination) connector for gantry (back) doors

2.1.4 Test equipment

- Quality phantoms, delivered with the system

2.1.5 Consumable items (lubricant, filters)

Item	Item number	Amount	Required
Air filter for PDC	30 68 095	1	1
Air filter for dehumidifier	27 93 250	5	1
Air filter set for IRS	73 95 325	1	1
Contains plastic frames and filters (small and large) for all 4 computers (IRS 1 and 2) or 4x 3 filter pads (IRS 3)	(for IRS 1) 73 95 473 (for IRS 2 / 2c / 2F) 83 79 344 (for IRS 3)		

Item	Item number	Amount	Required
Power brush block (UDC voltage) * complete, including brushes, screws and adjustment pins.	73 95 135	1	1
Control & low power brush block* complete, including brushes, screws and adjustment pins.	73 95 150	1	1
Stator ring brush (not for Sensation 64/Cardiac 64 and Open)	73 95 101	1	1
Lubricant Isoflex Topas NCA 52	15 90 863	1 kg	n.a.

* For convenience and fast service, complete brush blocks are available.

However, single brushes can be replaced during maintenance as well. (Go to [Spare parts / p. 13](#)) for more information)

2.1.6 Spare parts

Item	Item number	Amount	Required
UDC Power brushes Set of 20 UDC power brushes including 2 caps	73 95 143	1	1
Low power carbon brushes* Set of 12 low power brushes, including 2 caps	73 95 176	1	1
Control data unit* includes 16 brushes	73 95 168	1	1

* For convenience and fast service, two complete brush blocks are available. (Go to [Consumable items \(lubricant, filters\) / p. 12](#)) for more information)

However, it is also possible to replace single brushes during maintenance as well.

2.1.7 Cleaning materials

- Soft cloth for monitor screen
- Cleaning agent for system components (e.g. Plexiglas cleaner, mild soap , or commercially available cleanser)

2.2 Evaluating Reports

PM Checking system reports

NOTE

If remote access to the system (RD) is available, this action should be performed prior to the maintenance call!

To check for errors, the **Reports** | function can be used to obtain a readout of the logbook:

1. Switch on the system, go to “Local Service” and start the service software.
2. Start | Reports | in the upper navigation frame.
3. For additional information on how to handle the Eventlog, click the | **Help** | button in the upper navigation frame. Check the following reports:
 - **Tune up/QA/tests --> Tune up --> Table generations --> Defective channels**
Click the results and check the “Table Editing History” and “Affected Modules” if new defective channels were detected by the system and added. Refer to “Trouble shooting guide Gantry” (DMS) and change ADM-Modules if necessary.
 - **Performance --> Statistics**
Check results. If the system displays abnormal behavior (e.g. a high number of “Restarts”), check the logbook for more details. The “Error Hitlist” may also be helpful for further investigation.
 - **Performance --> Error Hitlist**
Check for errors that occur frequently. Follow the instructions in the description text for further actions.
 - **Tube --> Current History**
Check the tube history for tube arcings. If necessary, get the tube or continue with “Tube Errors”.
 - **Tube --> Errors**
Check for errors that occur frequently. Follow the instructions in the description text for further actions.

2.3 Evaluating the logbook

PM Error statistics readout

NOTE

If remote access to the system (RD) is available, this action should be performed prior to the maintenance call!

Read out the logbook and check for errors:

1. Switch on the system; go to “Local Service” and start the service software.
2. Start the **| event log |** in the upper navigation frame.
3. Check the event log for errors. For additional information on how to handle the event log, click the **| Help |** button in the upper navigation frame. If necessary discuss the results with the customer
 - Depending on the results, discuss the system performance with the customer.
 - Determine what, if any, concerns the customer has with respect to the system and how they can be addressed.
 - Note any customer concerns and the actions taken to resolve them in the Comments section of the Maintenance Certificate.
4. A statistical evaluation of relevant errors can be performed using the **| Reports |** function.

3 Safety Inspections

3.1 General

WARNING

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Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

⇒ The guidelines contained in this document have to be strictly followed.

3.1.1 Preparations

NOTE

Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained.

3.1.2 Required tools and auxiliary materials

Not required.

3.2 Function Tests

To perform the following function tests, the system has to be switched on and in working condition.

3.2.1 Measuring the patient leakage current (DIN VDE 751-1)

SI Patient leakage current checked

Note :

This measurement is required in Germany only, or in parts of German-speaking countries (e.g. Austria, Switzerland). Since no direct measurements are required for this system in accordance with DIN VDE 751-1, a justification is provided for the sake of completeness. For this reason, the justification is provided in German only:

System leakage current: not required.

Begründung: Festanschluß und bauseitige Schutzmaßnahmen bei indirektem Berühren (z.B. nach DIN VDE 0107)

patient leakage current at the part of the table being used (type B): not required

Begründung: Trennung zwischen spannungsführenden Teilen und Anwendungsteil durch schutzleiterverbundene Metallteile nach IEC 60601-1 , Abschnitt 17.a.2

patient leakage current at the ECG module (type BF) integrated into the table: not required

Begründung: Konstruktive Maßnahmen und Festeinbau

3.2.2 Checking the protective ground conductor resistance

This test checks the protective ground conductor resistance of the system.

NOTE

During maintenance several of the protective ground conductors will be handled. As a result, the protective conductor resistance measurement will be performed in the last chapter of this document. Refer to [\(Testing the protective ground conductor / p. 53\)](#) for more information.

3.2.3 Checking the function of the door contact switch

SI Door contact switch checked



This test checks whether a door switch blocks radiation.

NOTE

Omit this test if a door switch for the examination room has not been installed.

- Select any scan mode with radiation. To minimize radiation, select a mode with low dose, small slice settings.
- Press **Start** for the measurement and open the scan room door while radiation is ON.
⇒ The test is successful if the scan is blocked and a message appears.

3.2.4 System: Checking the radiation preparation indicator

SI Radiation preparation indicators checked

This test checks the function of the installed radiation preparation indicators.

NOTE

Omit this test if a radiation preparation indicator (usually a yellow warning lamp) has not been installed.



- Select any scan mode with radiation. To minimize radiation, select a mode with low dose, small slice settings.
- Load any mode and wait until the system prompts you to press the **Start** button. The radiation preparation indicators have to be ON.
- Press **Start** for the measurement.
- The indicator has to be ON during the scan.

3.2.5 System: Checking the radiation indicator

SI Radiation indicators checked



This test checks the function of the installed radiation "ON" indicators.

NOTE

This test is intended to check the radiation indications of the system. If additional (optional) radiation indicators (radiation warning lamps) were installed, they have to be checked accordingly.

- As described above, the system is still in scan preparation. (Any scan mode with radiation was previously selected).
- Press **Start** for the measurement and check whether
 - the acoustic radiation indicator in the control box is audible.
 - all radiation warning lamps (gantry front and back, control box) are ON were ON while radiation is applied.

3.2.6 System: Checking the radiation monitor (108%)

SI Radiation switch-off checked



This test checks the function of the X-Ray radiation monitor.

NOTE

The **TIMEOUT TEST** checks whether the radiation monitor in the generator is functioning properly. Load a scan mode with a higher number of readings than required for the respective scan time. After a radiation period of 110% (nominal scan time plus 10%), the generator must switch off radiation and transmit an error message.

- Open the test platform the for X-ray - Timeout test under **Test Tool --> Controller --> XRS --> XRay - Timeout** .
- Start and evaluate one test for each mode (topo, sequence and spirals).
 - Topogram
 - Sequence
 - Spiral (select “min” and “max” for measurement)
- ⇒ The test is successful, if the timeout value is < **110%** .

3.2.7 System: Checking the emergency STOP circuit

SI Emergency STOP circuit checked

This test checks the function of the emergency STOP circuit.

NOTE

This test checks whether all motor-driven system movements are immediately stopped when the **EMERGENCY STOP** button is actuated.

- Select any scan mode with radiation and rotation. To minimize radiation, select a mode with low dose, small slice settings.
- Press the red **EMERGENCY STOP** button on the keyboard while the gantry is rotating.
 - ⇒ This should stop all motor-driven system movement (gantry tilt, gantry rotation, table movement)
 - ⇒ The system is now in STOP status.
 - ⇒ A window appears with the request to <continue>.
- Confirm the pop-up window to continue measurement.
- Reload the mode and repeat the above test procedure for the **EMERGENCY STOP** button on all gantry control panels (front and rear).

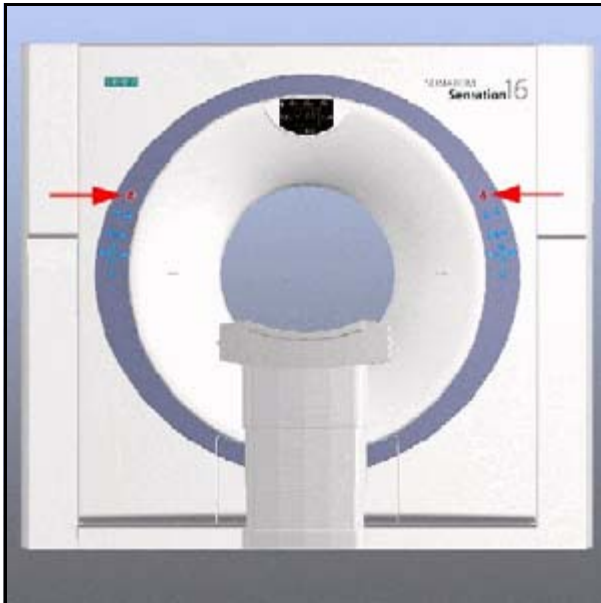


Fig. 1: Gantry front: location of E-Stop buttons

3.2.8 Gantry: Checking the tube cooling hoses

SI Tube cooling hoses checked

CAUTION

[hz_serdoc_F13G02U01M01]

Temperature of X-ray tube and/or tube cooling device parts may be above 70°C (with STRATON up to 130°C).

Risk of burns!

- ⇒ Avoid contact with oil and exposed parts of X-ray tube and cooling device.
- ⇒ Wait until X-ray tube and cooling device parts are chilled down.

CAUTION

[hz_serdoc_F04G05U01M04]

Injury from hot cooling oil may result, if cooling hoses become damaged.

Risk of injury!

- ⇒ Check the correct position of the cooling hoses and couplings as described in the maintenance instructions.

- Check the hoses of the tube cooling system
 - ⇒ They should not be loose or improperly routed
- Inspect all cooling hoses and cooling devices for oil leaks.

3.2.9 PHS: Checking the safety switches

SI Safety limit switches checked

This test checks the function of the PHS safety switches.

NOTE

Safety switches stop horizontal table travel when the foot-end top support makes contact with an obstacle or when someone steps on the floor-mounted connecting frame between the gantry and the table.

Activate the following safety switches:

- **PHS foot-end safety switch** by pressing the top support cover located at the foot-end toward the gantry
- **Floor-mounted connecting frame (PHS <--> Gantry) safety switch** by stepping on the black connecting frame on the floor between PHS and the gantry.

In each case, you will hear a distinct click. The GPC disables all push buttons for horizontal movement on the gantry panels. The corresponding push buttons are dimmed.

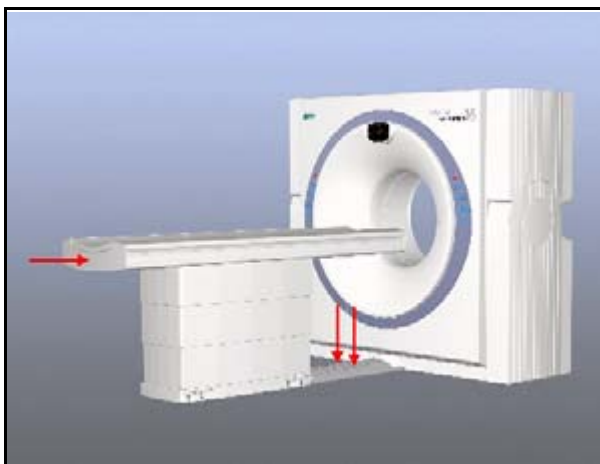


Fig. 2: PHS: location of safety switches

3.2.10 PHS: Checking the spindle brake

SI Function of spindle brake checked

This test checks the function of the PHS spindle brake.

- Enable the service functions by clicking: **Option --> Local Service .**
- Go to **Test Tool --> Controllers --> PTV**
- Select **Spindle brake** and press **<GO>** to start the test.
- Follow the test results on screen.

3.2.11 PHS: Checking the motor brake

SI Function of motor brake checked

This test checks the function of the PHS motor brake.

- Enable the service functions by clicking: **Option --> Local Service** .
- Go to **Test Tool --> Controllers --> PTV**
- Select **Motor brake** and press **<GO>** to start the test.
- Follow the test results on screen.

3.2.12 PHS: Checking the patient positioning aids

SI

Patient positioning aids checked

NOTE

This section tests the patient positioning supports attached to the patient table.

- Inspect the patient positioning accessories for damage (i.e. head cradle, holder, pads).
- Check the accessory bracket at the head-end of the table for damage.
- Check the lock-in mechanism on the patient positioning accessories.
 - Attach them and remove them from the accessory bracket at the head-end of the table several times.
 - Test criteria: the accessories should be safely and securely positioned on the patient table.

NOTE

If the locking mechanism does not function properly, determine the cause of the problem (i.e. spring, knob, or bracket insert in table) and replace the appropriate part(s).

3.2.13 PDC: Protection functionality test

SI

Circuit breaker checked

This test checks the function of the main circuit breaker F99.

1. Verify that F1 and F9 breakers in the PDC are switched on.
 - ⇒ If fuse F9 is switched off, the main breaker F99 will not work.
2. Switch on the power using the onsite ON/OFF switch. Make sure Somaris was shut down on the Navigator and Wizard.
3. Verify the status of the K100 and K101 LED"s ([Tab. 1 / p. 23](#)) .
4. If one of the LED"s has the wrong status, switch the power off and on using the onsite ON/OFF switch. If the problem persists, check all wiring
5. **Short Circuit Test**

If the status of the LEDs is o.k., press the S100 test key ([3/Fig. 3 / p. 23](#)) and hold for at least 2 seconds.

- ⇒ The main breaker F99 must trip off to the middle position.

6. To switch on the main breaker F99, first move the breaker to the OFF position and then to the ON position (requires extended pressure).
7. Verify the status of the K100 and K101 LEDs (Tab. 1 / p. 23) .
8. **Leakage current test**
 If the LED's status is o.k., press the S101 test key (4/Fig. 3 / p. 23) and hold for at least 2 seconds.
 ➡ The main breaker F99 must trip off to the middle position.
9. To switch on the main breaker F99, first move the breaker to the OFF position and then to the ON position (requires extended pressure).

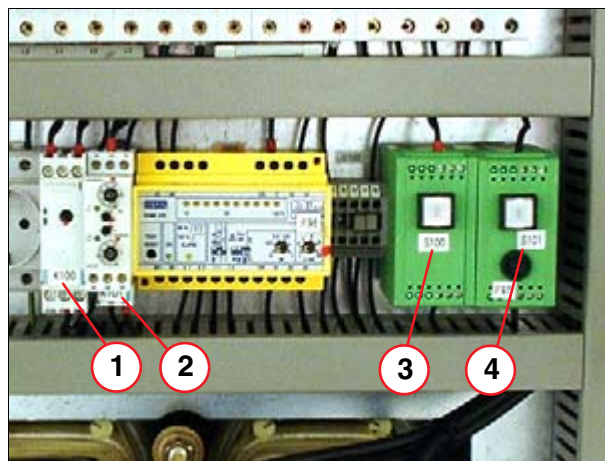


Fig. 3: K100 and K101

- Pos. 1 K100
 Pos. 2 K101
 Pos. 3 S100
 Pos. 4 S101

Tab. 1 LED's K100/K101

LED	green (230 V)	yellow	yellow
K100 Time delay relay	n.a.	OFF	OFF
K101 Current control relay (DCC T1)	ON	ON	n.a.

3.3 Next steps

NOTE

Use the "Maintenance Protocol" provided to document all work performed on the system during this part of the maintenance as well as the results obtained. Depending on the maintenance schedule, you may now conclude your maintenance activities with the concluding test or proceed with additional maintenance work according to the schedule (Recommended [\(Preventive maintenance \(part 1\) / p. 25\)](#)).

3.3.1 Concluding test

Q Condition of the CT system checked

- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

4 Preventive maintenance (part 1)

4.1 General

WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

⇒ **The guidelines contained in this document have to be strictly followed.**

The work steps in this section may not include all of the necessary details. If you require more detailed instructions (e.g. on how to remove the table covers), please refer to the latest version of the service documentation "PHS - Replacement of parts" - CT02-023.841.03.xx.xx, for further details.

4.1.1 Preparations

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the 'Maintenance Protocol' provided.

- Move the table to the lowest vertical position.
- Remove
 - the center mounting screw of the base cover (front side)
 - the center mounting screw of the base cover (back side)
- Move the table to maximum height.
- Switch off the system.



⚠ WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- ⇒ **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- ⇒ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- ⇒ **Ensure via measurements that all voltages are switched off.**
- ⇒ **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

4.1.2 Required tools and auxiliary materials

- Spring scale
- Phantom holder, no. 73 96 943, delivered with the system
- Standard service kit
- Lubricant TOPAS NCA 52 (item number: 15 90 863)

4.2 Patient handling system

4.2.1 Lubricating the spindle nut

PM Spindle nut lubricated

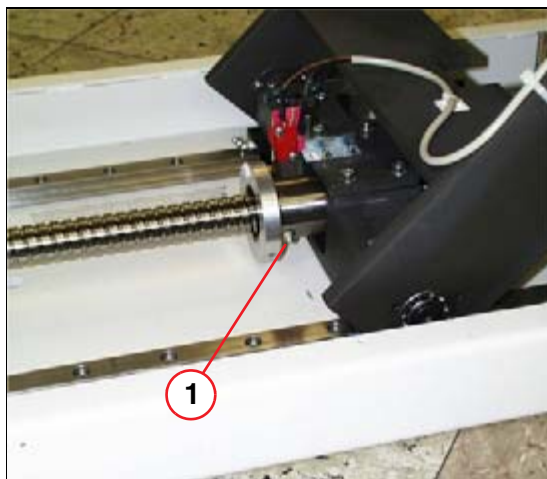


Fig. 1: Spindle: location of grease nipple
Pos. 1 Grease nipple

- Place the connector of the lubricant gun on the lubrication fitting of the spindle nut (1/ Fig. 1 / p. 27).
- Pump lubricant into the spindle nut until lubricant flows out of both sides.
- Clean any old lubricant from the spindle.
- Remove the connector of the lubricant gun from the spindle nut.

4.2.2 Lubricating the linear guides on the lower part of scissor mechanism

PM Lower part of the scissor mechanism: Linear guides lubricated

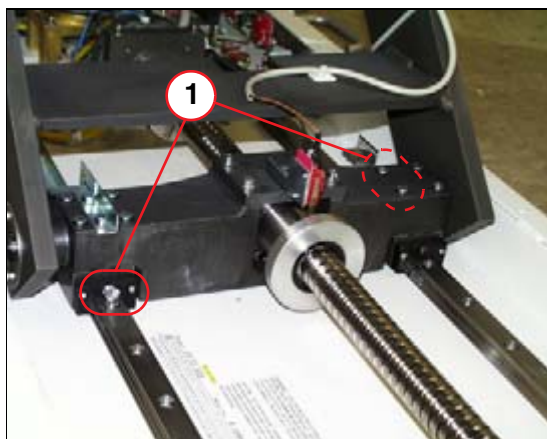


Fig. 2: Scissor: location of grease nipple
Pos. 1 Location of grease nipple

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/ Fig. 2 / p. 27).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.3 Lubricating the linear guides of the upper part of scissor mechanism

PM Upper part of the scissor: Linear guides lubricated

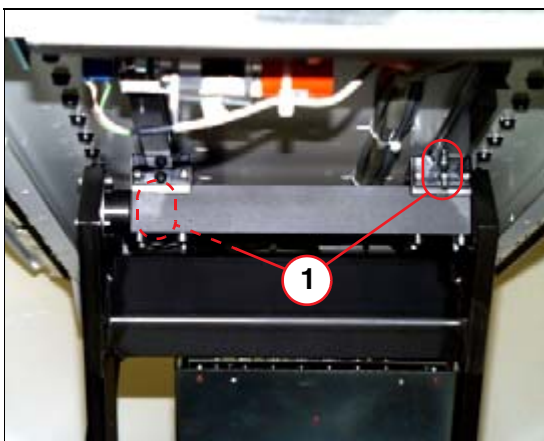


Fig. 3: Scissor: location of grease nipple
Pos. 1 Location of grease nipple

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/ Fig. 3 / p. 28).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.4 Lubricating the guide rails of the tabletop

PM Tabletop: Guide rails lubricated

- To access the guide rails, remove the cover(s)
 - under the tabletop at the head-end of the table
 - at the foot-end of the table
- Move the tabletop accordingly to access the grease fittings shown.

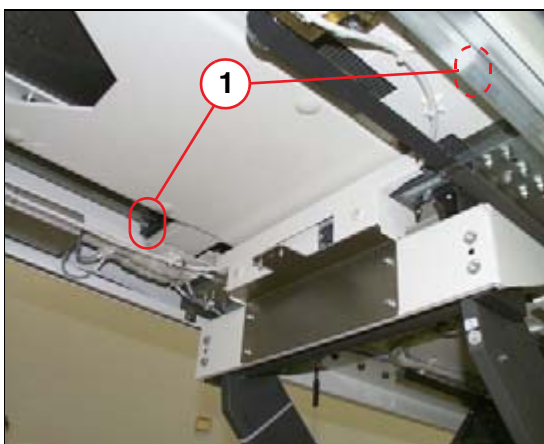


Fig. 4: Tabletop: location of grease nipple
Pos. 1 Grease nipple

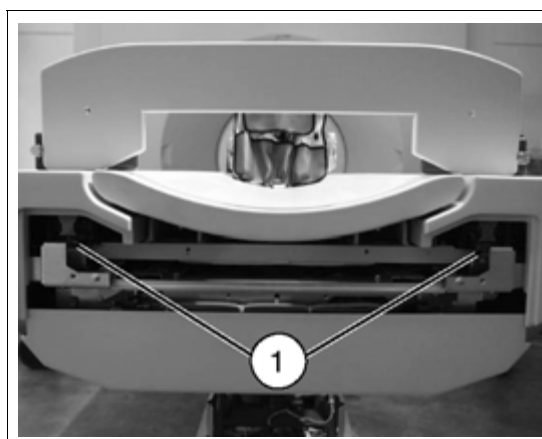


Fig. 5: Linear guides of the tabletop (foot-end of table)

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/ Fig. 5 / p. 28).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.5 Lubricating the guide rails of the top support

PM Top support: Guide rails lubricated

- Move the tabletop accordingly to access the grease fittings shown.

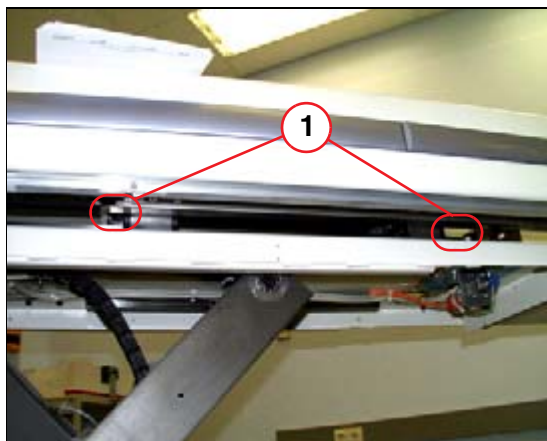


Fig. 6: Table: location of grease nipple
Pos. 1 Grease nipple

- Place the connector of the lubricant gun on one of the lubrication fittings of the guide rail to be lubricated (1/Fig. 6 / p. 29).
- Pump lubricant into the carriage until the new lubricant is visible on the guide rail.
- Repeat the procedure for the second guide rail.
- Clean any old lubricant from the guide rails with a soft cloth.

4.2.6 Checking the movement of the tabletop

PM Movement force for horizontal tabletop travel checked

- Insert the phantom holder delivered with the system into the socket on the tabletop.
- Secure the phantom holder to the tabletop using a cable tie.
- Move the tabletop toward the gantry (approximately 100 mm).
- Move the top support of the table toward the gantry (approximately 100 mm).
- Hook the spring gauge into the cable tie.
- Using the spring gauge, pull the tabletop toward the foot-end of the table. Maintain a constant force.
 - View the spring gauge to check the force required to move the tabletop.

NOTE

The force required to move the tabletop should be < 160 N.
--

4.2.7 Checking the movement of the top support

PM Movement force for horizontal top support travel checked.

- Pull the tabletop toward the foot-end until it reaches the end stop. (The top support is still positioned about 100 mm toward the gantry)
- Using the spring gauge, pull the top support toward the foot-end of the table. Ensure that the force remains constant.
 - View the spring gauge to check the force required to move the tabletop.

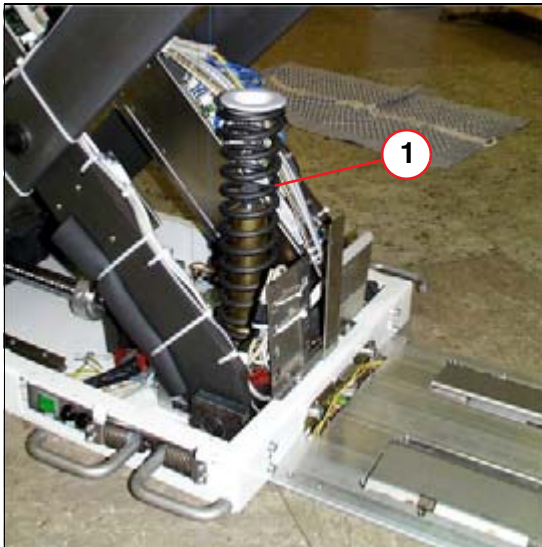
NOTE

The force required to move top support should be < 160 N.

- Remove the cable tie, the phantom holder, and the spring scale.

4.2.8 Checking the compression spring on the patient table

PM Compression spring checked



- Check the spring for mechanical damage such as cracks or breakage.

Fig. 7: Location of compression spring
Pos. 1 Compression spring

4.2.9 Cleaning the motor brake of the couch lift motor

PM Vertical motor brake cleaned

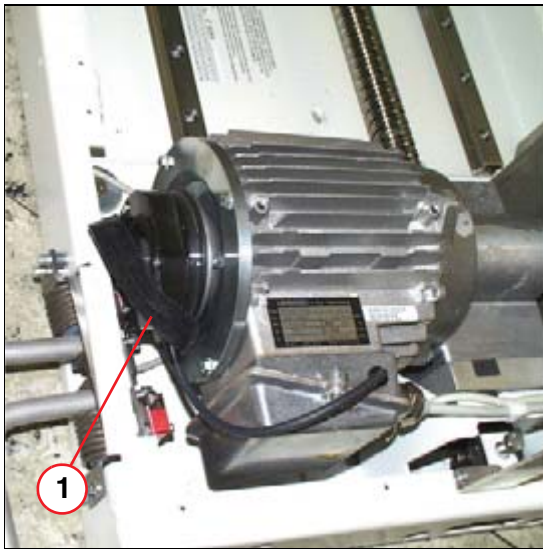


Fig. 8: *PHS vertical motor*
Pos. 1 Location of the rubber gasket

- Pull off the rubber gasket (1) from the vertical motor brake.
- Tap the vertical motor brake lightly or switch the drive on and off briefly to dislodge the abrasion dust from the motor brake.
- Clean the motor brake carefully and remove dust using a vacuum cleaner.
- Reinstall the rubber gasket (1) on the vertical motor brake

4.3 Next steps

- Reattach all covers on the PHS.

If necessary, use the service documentation, “PHS- Replacement of parts” - CT02-023.841.03.xx.xx, for further details.

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the 'Maintenance Protocol' provided. Depending on the maintenance schedule, you may either conclude the maintenance activities with the concluding test or proceed with additional maintenance work according to the schedule (Recommended [\(Preventive maintenance \(part 2\) / p. 33\)](#)).

4.3.1 Concluding test

Q Condition of the CT system checked

- Switch on the system.
- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

5 Preventive maintenance (part 2)

5.1 General



[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

⇒ **The guidelines contained in this document have to be strictly followed.**

Perform the maintenance workflow according to this document.

The work steps in this section may not include all of the necessary details. If you require more detailed instructions (e.g. how to remove the gantry covers or the gantry tunnel), please refer to the latest version of the service documentation "Gantry - Replacement of parts" - CT02-023.841.01.xx.xx, for further details.

5.1.1 Preparations

- Switch on the system.
- Move the table to the lowest position.
- Open the front door of the gantry.
- Disable gantry rotation via service switch S301.

5.1.2 Required tools and auxiliary materials

- Standard service kit
- Sliding calipers
- Lubricant: Isoflex Topas NCA 52 (15 90 863)
- Grease gun

5.2 Gantry

5.2.1 Functional check of the fans in the generator electronic box

SI Function of E-box fans checked



Fig. 1: Generator electronic box

- Check whether the 4 fans of the generator electronic box are working.

5.2.2 Functional check of the HV transformer fans

SI Function of HV transformer fans checked

Check whether both fans located on the HV transformer are working.

5.2.3 Functional check of the C-box fans (only Sensation 10 / 16 / Cardiac)

SI Function of C-box fan checked

Check whether the C-box fan is working.

5.2.4 Functional check of the DMS fans

SI Function of DMS fans checked

Check whether both DMS fans are working.

5.2.5 Cleaning the slip ring compartment

PM Slip ring compartment and assembly cleaned



- Switch off the system.

WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- ⇒ Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).
- ⇒ Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).
- ⇒ Ensure via measurements that all voltages are switched off.
- ⇒ Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.

- Remove the back gantry cover.
Make sure to disconnect X350 and PE350 prior to removing the back cover completely!
- Remove the gantry funnel.
Make sure to disconnect X312 A-D.
- Remove carbon dust from the
 - complete slip-ring compartment
 - slip ring assemblyusing a vacuum cleaner. **Do not use alcohol or other cleaning liquids.**

5.2.6 Cleaning/checking the power brush block

PM Power brush block cleaned

PM Power carbon brushes checked

PM Power carbon brushes replaced

NOTE

Perform this section every 6 months.

Omit this section if you have to replace the complete power brush when performing "Full maintenance" (every 12 months)

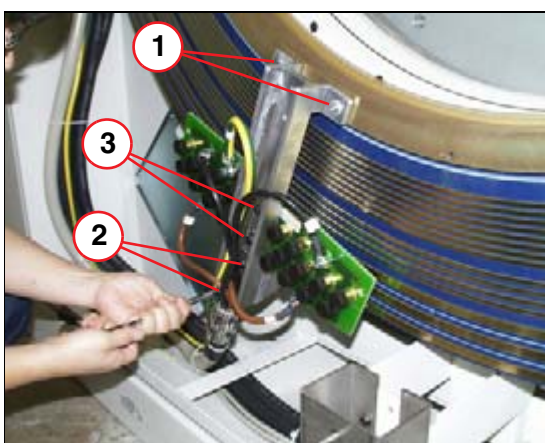


Fig. 2: Power brush assembly

Pos. 1 Support bracket: location of inner fastening screws

Pos. 2 Support bracket: location of outer fastening screws

Pos. 3 Brush block: location of mounting screws

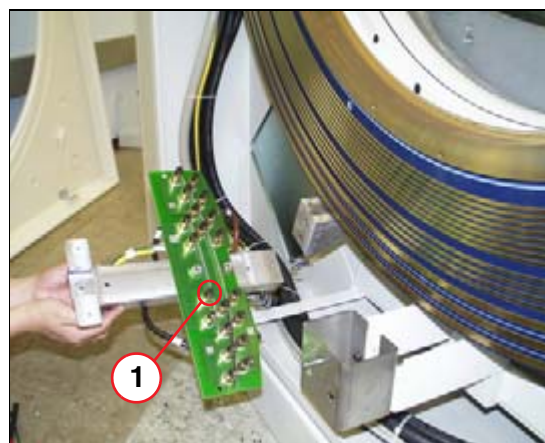


Fig. 3: Power brush assembly

Pos. 1 Check the each carbon brush (20 pieces)

- Remove the four mounting screws (two inner and two outer screws) of the power brush assembly and pry back the assembly. Be careful when handling the brush block; carbon brushes may become damaged!
- Remove the caps from the power carbon brushes and carefully pull out each brush. Put all brushes aside in a safe place.
- Use a vacuum cleaner to remove carbon dust from the power brush assembly. If necessary, you may use a clean soft-bristled toothbrush. Make sure that each brush guide is cleaned well to ensure functionality.
- Each carbon brush has to be **checked or replaced** if
 - any damage is visible (e.g. scratched or broken surface)
 - its length is out of tolerance. Each carbon brush has an engraved triangle which indicates the minimum tolerance.

NOTE

Ensure that the brush length will not go below the 12 mm limit prior to the next regular maintenance call.

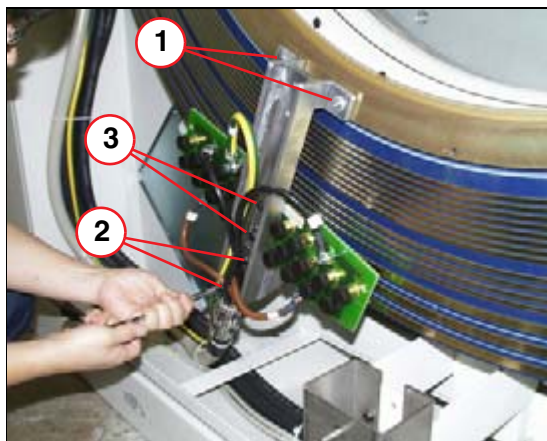
- Reinstall all power brushes. Install all caps. Make sure all brushes are movable.
- Reinstall the power brush assembly. Refer to (Tab. 1 / p. 36)

Tab. 1 Information regarding screw torque

Screw	Tightening torque	Loctite 243
(1/Fig. 2 / p. 36)	3 Nm	yes
(2/Fig. 2 / p. 36)	24 Nm	no

5.2.7 Replacing the power brush block

PM Power brush block replaced

NOTE**Perform this section every 12 months. (Full maintenance)***Fig. 4: Power brush assembly*

- Pos. 1 Support bracket: location of inner fastening screws
- Pos. 2 Support bracket: location of outer fastening screws
- Pos. 3 Brush block: location of mounting screws

- Loosen all cable connections on the power brush block.
- Loosen the two mounting screws of the power brush block (about 1-2 turns).
- Remove the four mounting screws (two inner and two outer screws) of the power brush assembly and carefully pry back the assembly.
- Loosen the two mounting screws of the power brush block completely and remove the block.
- Install the new power brush block. Do not tighten the screws completely.
- Reinstall the power brush assembly by fastening the 4 screws. Refer to [\(Tab. 2 / p. 37\)](#)
- Move the gantry by hand until the two adjustment holes of the slip ring are aligned with the brush block. A 10mm pin may be helpful for this adjustment. Align the brush block and secure it with 2 screws. Refer to [\(Tab. 2 / p. 37\)](#)
- Connect all cables to the power brush block.

Tab. 2 Information regarding screw torque

Screw	Tightening torque	Loctite 243
(1/Fig. 4 / p. 37)	3 Nm	yes
(2/Fig. 4 / p. 37)	24 Nm	no
(3/Fig. 4 / p. 37)	6 Nm +/- 10%	yes

5.2.8 Checking/replacing the voltage and control data brushes

PM Voltage and control data brushes checked

PM Voltage and/or control data brushes replaced

NOTE

Perform this section every 6 months.

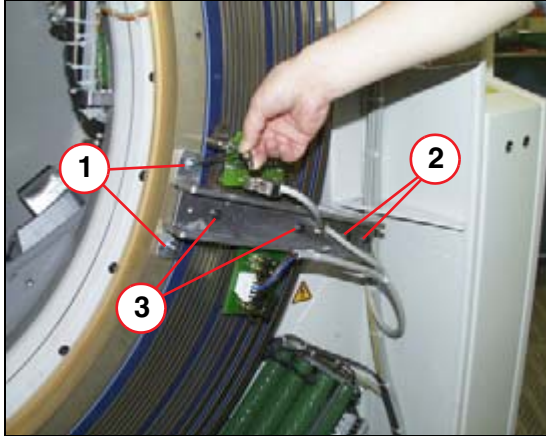


Fig. 5: Brush block for low power & control data

- Pos. 1 Support bracket: 2 x inner fastening screws
- Pos. 2 Support bracket: 2 x outer fastening screws
- Pos. 3 Brush block: 2 x fastening screws

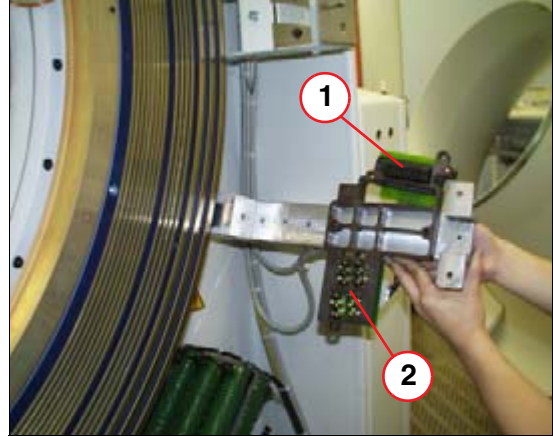


Fig. 6: Brush block for control data & low power

- Pos. 1 Control data brushes
- Pos. 2 Low power brushes

- Disconnect the X1 and X2 connectors.
- Loosen the two mounting screws of the power brush block (about 1- 2 turns).
- Remove the four mounting screws (two inner and two outer screws) and carefully pry back the assembly. Do not lose the plastic washers.
- Check the length of the brushes.

NOTE

Minimum length of carbon brushes required:

Control data brushes - 8mm

Low power brushes - 14mm

Ensure that the brush length will not fall below the minimum length before the next regular maintenance call.

- Replace the brushes if necessary.

NOTE

For convenience and fast service, complete brush blocks should be replaced. However, single brushes can be replaced during maintenance as well.

- Loosen the two mounting screws of the power brush block completely and remove the block.
- Install the new power brush block. Do not tighten the screws completely.

- Reinstall the power brush assembly. Refer to (Tab. 3 / p. 39)
- Move the gantry by hand until the two adjustment holes of the slip ring are aligned with the brush block. An 8 mm pin may be helpful for this adjustment. Align the brush block and secure it with 2 screws.
- Connect the X1 and X2 connectors to the brush blocks.

Tab. 3 Information regarding screw torque

Screw	Tightening torque	Loctite 243
(1/Fig. 5 / p. 38)	3 Nm	yes
(2/Fig. 5 / p. 38)	28 Nm	no
(3/Fig. 5 / p. 38)	6 Nm +/- 10%	yes

5.2.9 Replacing the stator ring brush (only Sensation 10 / 16 / Cardiac)

PM Stator ring brush replaced

NOTE

Perform this section every 6 months.

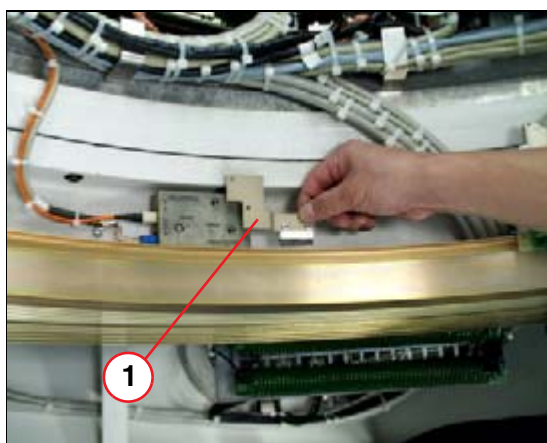


Fig. 7: Replacement of stator ring brush

Pos. 1 Stator ring brush

- Turn the rotating part of the gantry and position the transmitter of the data transmission line in the 6 o'clock position.
- Remove the 2 mounting screws of the stator ring brush.
- Insert the new stator ring brush. Make sure that the brush has contact. As an indicator, the brush hairs should be slightly bent. Fasten the 2 screws.

5.2.10 Lubricating the rack and pinion gear of the UHR assembly (only Sensation 64, Cardiac 64)

NOTE

No lubrication is required on the UHR-Plus. It is maintenance free.

PM Rack and pinion gear lubricated (only Sensation 64, Cardiac 64, not UHR-Plus)**NOTE****Perform this section every 12 months (full maintenance)**

- Clean old lubricant off the gear rack by using a lint-free cloth.
- Lubricate the gear rack with grease Isoflex Topas NCA52, appr. 0.4 ccm should be used.
- Evenly distribute the grease with a brush all over the gear rack.

NOTE**Do not use the grease nipple for lubricating.****5.2.11 Cleaning/checking the detector window****PM Detector window cleaned/checked**

- Ensure that the radiation input window of the detectors is free from all contaminants such as contrast medium.
- Clean the detector window with alcohol, if necessary.
- Ensure that the radiation input window of the detector is not broken or severe scratched. This may impair the image quality of the system!
- Replace the radiation input window, if necessary.

5.2.12 Lubricating the main bearing**PM Main bearing lubricated****NOTE****Perform this section every 12 months. (Full maintenance)****NOTE****Approximately 15 grams of lubricant should be used. Determine in advance how much lubricant is released with each application of the grease gun, so you will know how much lubricant to apply.**

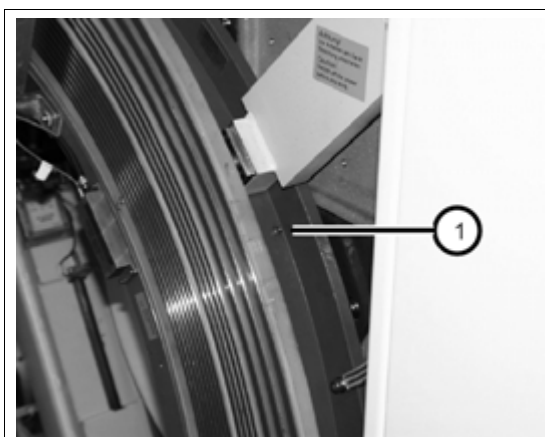


Fig. 8: Grease fitting, upper right side in the back of the gantry

- Connect the grease gun to the lubrication fitting (1/Fig. 8 / p. 41).

- Insert approximately 2.5 g of lubricant.
- Rotate the gantry by hand approximately 120 °.
- Insert another approximately 2.5 g of lubricant.
- Rotate the gantry by hand again approximately 120 °.
- Insert another approximately 2.5 g of lubricant.
- Rotate the gantry by hand several turns.
- Repeat the procedure.
- Disconnect the grease gun from the lubrication fitting.
- Check for and remove any excess lubricant at the bearing.
- Set service switch S301 to **ON**.

5.2.13 Replacing the filter in the dehumidifier

PM

Filter in Dehumidifier replaced



Fig. 9: Dehumidifier: filter replacement

- Remove the filter assembly from the dehumidifier (Fig. 9 / p. 41).
- Replace the filter.
- Reinstall the filter assembly in the dehumidifier.

5.3 Power distribution cabinet

5.3.1 Replacing the air filter

PM Air filter in PDC replaced

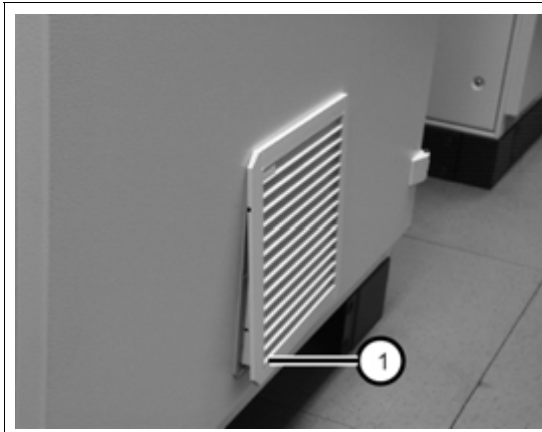


Fig. 10: PDC air filter

- Remove the air filter assembly from the front side of the PDC.
(1/ Fig. 10 / p. 42).
- Replace the filter.
- Reinstall the air filter assembly.

5.3.2 Checking/replacing the varistors and discharger

PM Varitors / discharger checked/replaced

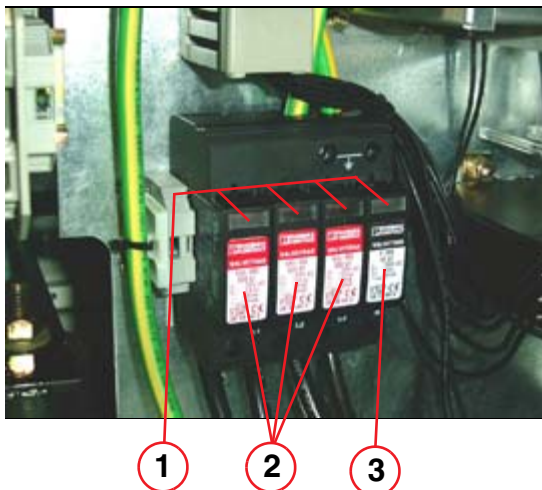


Fig. 11: Varistors and discharger

Pos. 1 Identification for defect
Pos. 2 Varistor
Pos. 3 Overvoltage protection

- A defective varistors / discharger will be displayed as “**defect**” in the small windows (pos. 1)
- If defective, replace it, as described in replacement of parts.

5.4 Water cooling system

5.4.1 Checking the water pressure

PM Water pressure checked

NOTE

Switch off the water cooling system to measure the water pressure.

- Place the manometer on the valve of the gantry cooling system.
- Open the front cover of the heat exchanger cabinet.
 - ⇒ **The nominal pressure for the cooling system is 2.0 bar +/- 0.2 bar.**
- Ensure the water pressure; if necessary, increase the water pressure by adding water. For detailed instructions, refer to service documentation "WCS- Replacement of parts"-CT02-023.841.05.xx.xx.
- Inspect the water pipes for leakage.

5.4.2 Cleaning the filter in the water-air cooling system

PM Filter cleaned

- Open the large tie wraps for the air hose at the air outlet (The hose closest to the rear of the cabinet, left or right side).
- Remove the hose from the outlet.
- Clean the filter using a vacuum cleaner.
- Connect the air hose and secure it using tie wraps.

For detailed instructions, refer to service documentation "WCS- Replacement of parts"-CT02-023.841.06.xx.xx.

5.5 Imaging system

5.5.1 Cleaning the ICS, IRS and IES

PM Air inlets of ICS, and IES cleaned

PM IRS air filters changed

- Remove or open the front covers of the ICS, IRS and IES.
- Change air filters of IRS 1 and IRS 2
- Or change air filters of IRS 3.

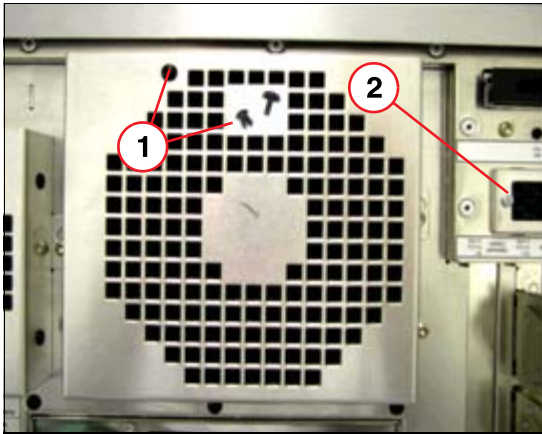


Fig. 12: Mounting of filter pads

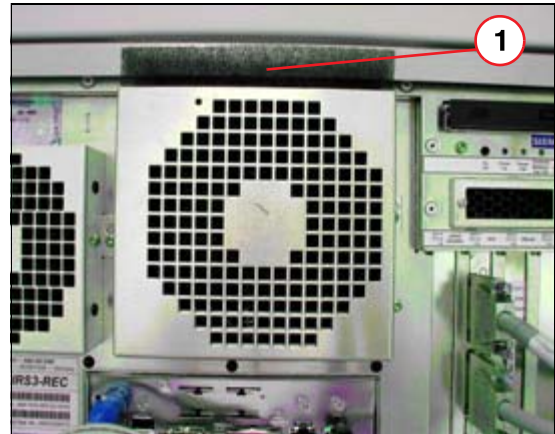


Fig. 13: Replacing the filter pad

- Remove the plastic, expanding rivet of the fan cover with a knife (1/Fig. 12 / p. 44). This rivet is only a transportation lock and must not be installed again.
- Push the filter pads upward using a screwdriver (1/Fig. 13 / p. 44) and remove it from the filter frame.
- Insert the new filter into the filter frame and push it down
- Loosen the Phillips screw (2/Fig. 12 / p. 44) on the frame for the small filter pad (111mm x 21 mm).
- Move the frame to the left and remove it from the rack.
- Replace the filter pad.
- Mount the frame with the new filter pad onto the computer rack and fix it with the Phillips screw.
- Clean the air inlets of the ICS and Wizard using a vacuum cleaner.

5.6 Next steps

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the 'Maintenance Protocol' provided. Depending on the maintenance schedule, you may either conclude the maintenance activities with the concluding test or proceed with further maintenance work according to the schedule (Recommended 'Image Quality Check' or, if necessary, first 'Preventive Parts Replacement').

5.6.1 Concluding test

Q Condition of the CT system checked

- Switch on the system and activate the service platform.
- Activate **Test Tools** --> **Rot Mode** and select the **slowest time** available.
- Press <Go> and rotate the gantry for approx. 1 minute to distribute the grease in the main bearing.
- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

6 Preventive parts replacement

6.1 General

WARNING

[hz_serdoc_F13G01U12M01]

Failure to observe the guidelines specified in this document may lead to serious injury or loss of life.

Risk of accident and/or injury!

⇒ The guidelines contained in this document have to be strictly followed.

6.1.1 Preparations

NOTE

All work performed on the system during this part of the maintenance as well as the results obtained have to be documented in the 'Maintenance Protocol' provided.

6.1.2 Required tools and auxiliary materials

- Standard service kit

6.2 Uninterrupted power supply

6.2.1 Prerequisites

Tools and auxiliary equipment

- Standard tool kit

Time and manpower

- 2 persons
- 1 hour

6.2.2 Replacing the standard UPS

PM UPS replaced

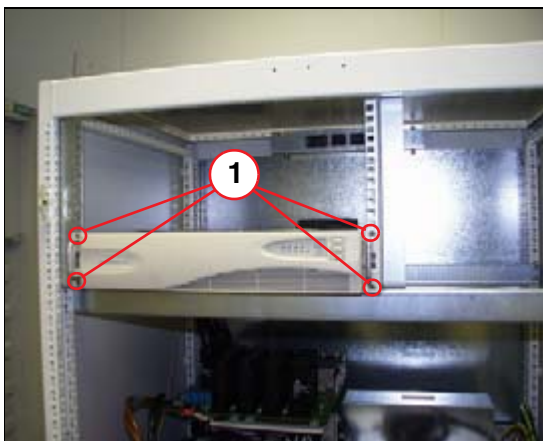


Fig. 1: UPS- uninterruptible power supply

Pos. 1 4 x fastening screws



- The system must be switched off completely prior to any intervention of the system.

WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- ⇒ **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- ⇒ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- ⇒ **Ensure via measurements that all voltages are switched off.**
- ⇒ **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

- Check the status LED's of the UPS. All LED's must be off!
- Remove the four mounting screws (1/Fig. 1 / p. 47).
- Carefully pull the UPS halfway out of the PDC frame.
- Disconnect the plugs on the back side of the UPS
 - 1 male plug from the UPS connector "LOAD1"
 - 1 female connector of UPS-1 from the UPS connector "INPUT"
- Pull the UPS carefully out of the PDC frame. **Due to the heavy weight, a second person is required.**
- Connect the new UPS control cable (delivered with the UPS) to the "COMM" port at the UPS.
- Check the locations of the power plugs on the PDC. It will be difficult to find the location when the UPS is already halfway in the PDC.
- Insert the new UPS halfway into the PDC frame.
- Connect the male plug of UPS-1 to the UPS connector "LOAD1".
- Connect the female connector of UPS-1 to the UPS connector "INPUT".
- Route the UPS control cable from "COMM" port at the UPS to the PDC-X270 .
- Push the UPS completely into the PDC frame.
- Secure the UPS with four Allen screws(1/Fig. 1 / p. 47).
- Switch on the UPS

6.3 Next steps

NOTE

The date of replacement as well as the date of next replacement has to be documented in the 'Maintenance Protocol' provided. Depending on the maintenance schedule, you may either conclude the maintenance activities with the concluding test or proceed with the maintenance according to the schedule (Recommended 'Image Quality Check').

6.3.1 Concluding test

Q Condition of the CT system checked

- Simulate customer usage to ensure that the system is in good working condition before turning it over to the customer.

7 Image quality - constancy

Q Image quality check completed

NOTE

The procedure for the image quality - constancy test is described in the “SOMATOM Operator Manual, chapter “Quality Assurance - Constancy Test”

8 Final steps

NOTE

You have completed the maintenance. All work performed on the system during maintenance and the results obtained have to be documented in the 'Maintenance Protocol' provided. After performing the "Final Steps," provide the customer with the completed protocol and, depending on organizational requirements, return a copy of it to your regional unit or field service office.

8.1 Installing the unit covers

SIM Covers and protective conductor cables are installed

NOTE

All metal system covers are equipped with a plug or screw connection for the protective conductor. Ensure that the protective conductor cable is connected to this point when the system covers are installed. The protective conductor cable is included with the components.

8.2 Testing the protective ground conductor

NOTE

The protective conductor resistance must be $\leq 100 \text{ m}\Omega$. Perform a measurement from the central protective ground terminal in the on-site power distributor to every conductive contact area of the CT components listed below.

- The system has to be switched off completely.
- Use protective conductor meter, e.g. P/N 4415899 RV090 or SECUTEST SIII, to perform the test.
- To ensure correct results, unroll the cable completely from the device prior to performing the test.

Measure protective conductor resistance for the following:

Note the equipment used in the maintenance protocol (type, manufacture, serial number and date of calibration).

SI Gantry: protective conductor resistance $\leq 100 \text{ m}\Omega$

Measure from the ground terminal to metallic parts on the gantry stand and gantry.

SI PHS: protective conductor resistance $\leq 100 \text{ m}\Omega$

Measure from the ground terminal to metallic parts on the table base and tabletop.

SI PHS: protective conductor resistance $\leq 100 \text{ m}\Omega$ **SI IRS cabinet: protective conductor resistance $\leq 100 \text{ m}\Omega$.**

Measure from the ground terminal to the metal housing.

SI PDC cabinet: protective conductor resistance $\leq 100 \text{ m}\Omega$.

Measure from the ground terminal to the metal housing.

SI WCS cabinet: protective conductor resistance $\leq 100 \text{ m}\Omega$.

Measure from the ground terminal to the metal housing.

SI ICS: protective conductor resistance $\leq 100 \text{ m}\Omega$

Measure from the ground terminal to the metal housing.

SI ICS monitor: protective conductor resistance $\leq 100 \text{ m}\Omega$.

Measure from the ground terminal to the metal housing.

- SI** **IMS (Wizard) : protective conductor resistance $\leq 100 \text{ m}\Omega$**
Measure from the ground terminal to the metal housing.
- SI** **IMS (Wizard) Monitor: protective conductor resistance $\leq 100 \text{ m}\Omega$**
Measure from the ground terminal to the metal housing.

8.3 Cleaning the monitors

SIM Monitors cleaned

- Clean the monitors.
- Remove any dirt or contamination caused by maintenance work.

8.4 Testing the system application

Q Condition of the CT system checked

- To ensure that the system is in good working condition before turning it over to the customer, simulate several system functions typical of the customer.

9 Changes to previous Version

Chapter	Change	Reason
2,5	New IRS 3	Change history 142692
5	New UHR-Plus	Change history 124842

H	hz_serdoc_F04G05U01M04	20
	hz_serdoc_F13G01U03M01	10
	hz_serdoc_F13G01U03M02	11
	hz_serdoc_F13G01U05M0310, 26, 35, 48	
	hz_serdoc_F13G01U12M019, 16, 25, 33, 46	
	hz_serdoc_F13G01U12M02	9
	hz_serdoc_F13G02U01M01	20
	hz_serdoc_F13G07U01M01	9